

AI for Medicine and Life Science - *AI for text and language data*

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2026-09-14



Module 1 content

Course intro

Foundations of AI recap

Language & speech data

AI applications related to language and speech

Foundations of AI recap

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What is AI?

Maths

What is artificial intelligence?

Artificial intelligence

Imitation of natural intelligence in computers

The Google logo is displayed in its standard multi-colored font (blue, red, yellow, blue, green, red) on a white background.

What are the symptoms of covid19|?



what are the symptoms of **covid 19** in dogs

What is artificial intelligence?

Artificial intelligence

Imitation of natural intelligence in computers

Where should we begin?



Ask anything



Think



What is artificial intelligence?

Artificial intelligence

~~Imitation of natural~~ intelligence in computers

Where should we begin?



Ask anything

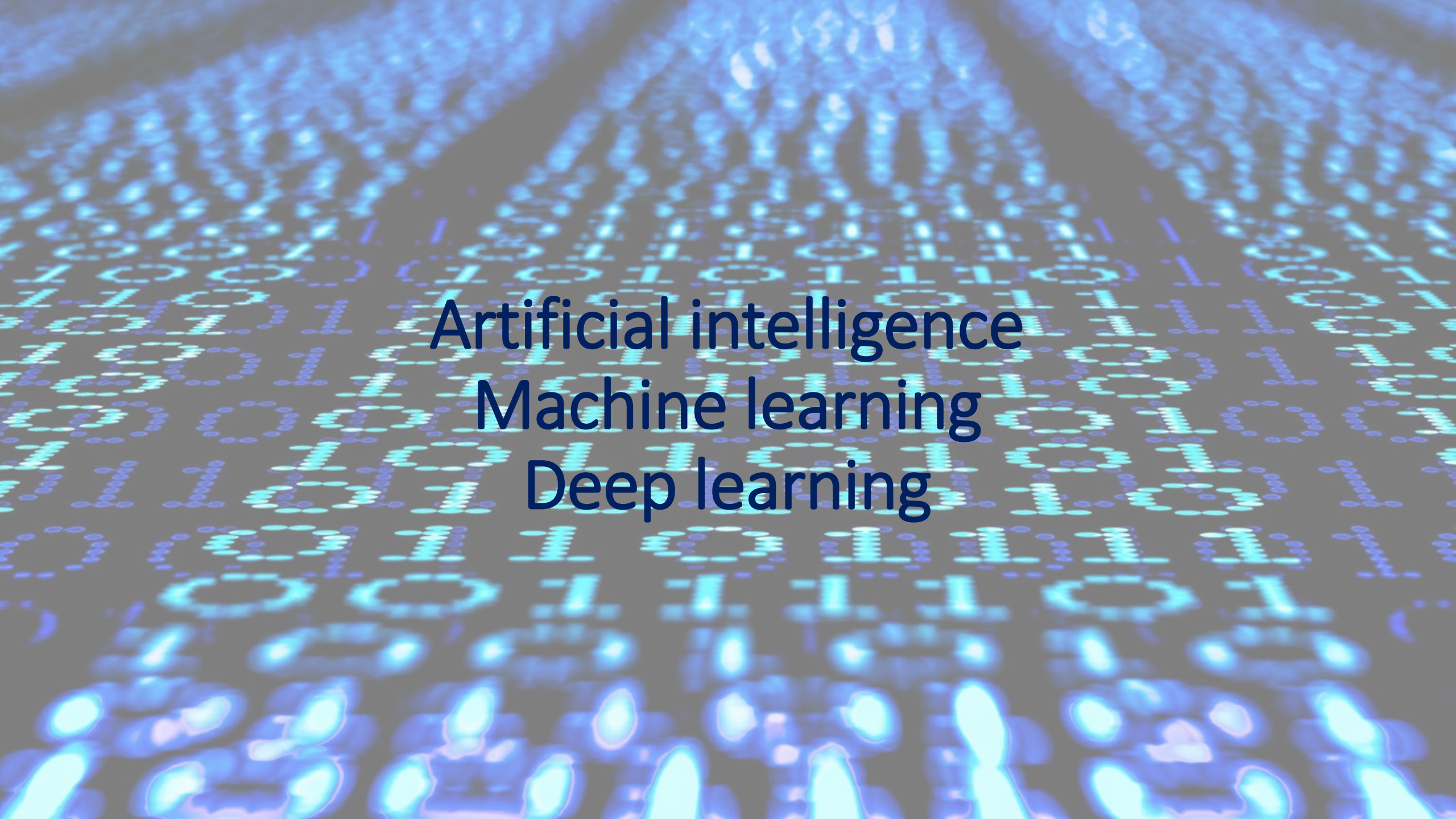


Think



Subdomains of AI that cover language and speech

- Natural language processing (NLP): Focuses on understanding, manipulating, and generating human language (mostly written language)
- BioNLP = NLP related to medicine/life sciences
- Speech processing: Focuses on processing the acoustic signals of spoken language



Artificial intelligence
Machine learning
Deep learning

What is artificial intelligence?

Artificial intelligence

Intelligence in computers

Machine learning

Computers learn decision rules from examples

Height	Movement	Decision
50 m	no	Plant
7 m	yes	Animal
10 cm	no	Plant
2 mm	yes	Animal
30 cm	yes	???

What is artificial intelligence?

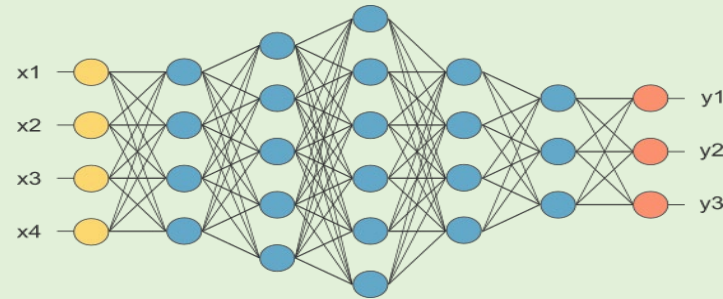
Artificial intelligence

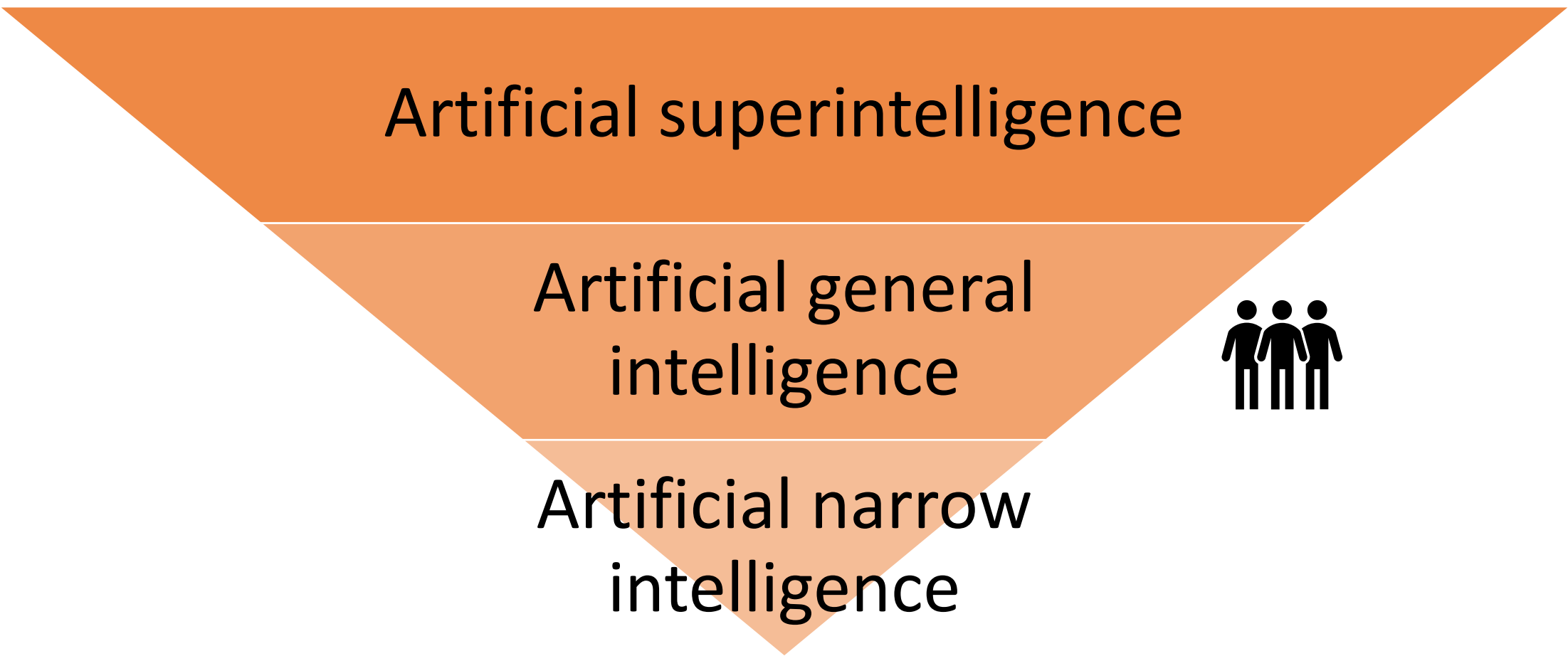
Intelligence in computers

Machine learning

Computers learn decision rules from examples

Deep learning



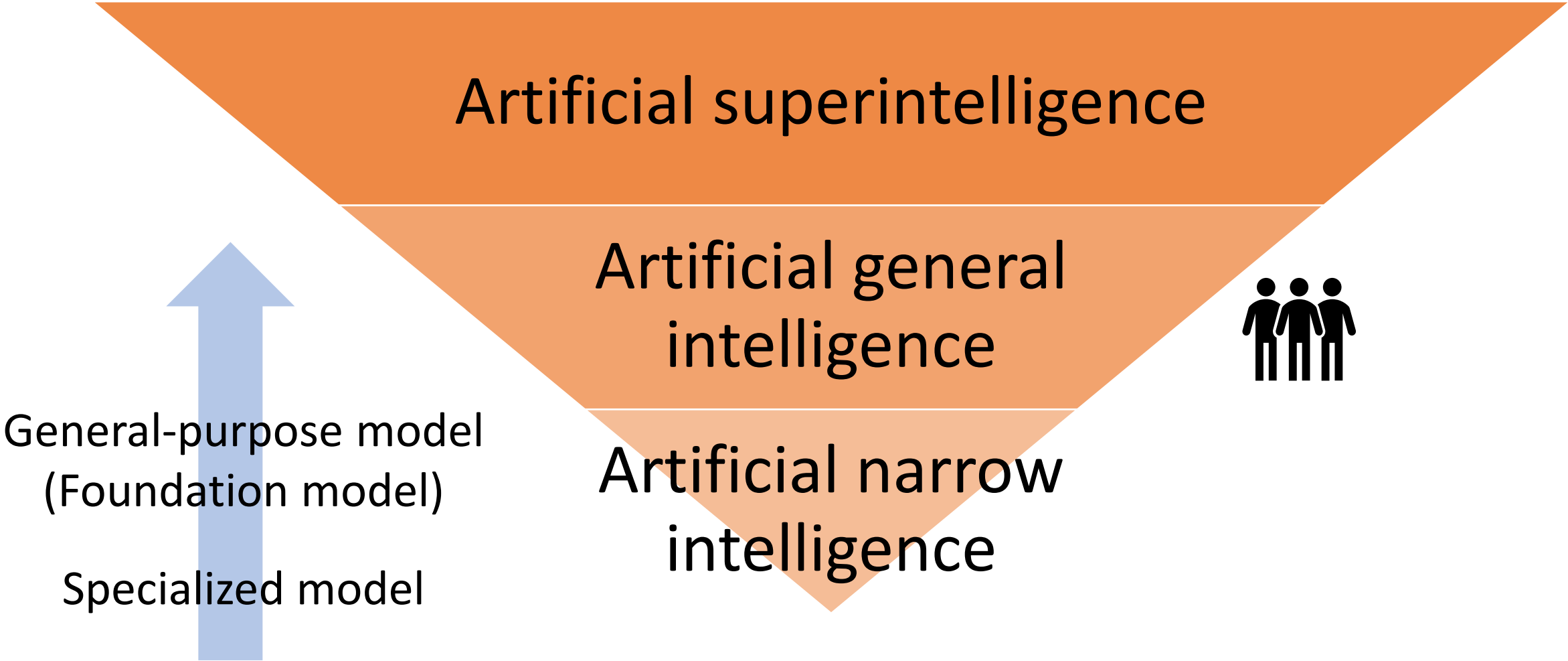


Artificial superintelligence

Artificial general
intelligence



Artificial narrow
intelligence



Artificial superintelligence

Artificial general
intelligence



Artificial narrow
intelligence

General-purpose model
(Foundation model)

Specialized model

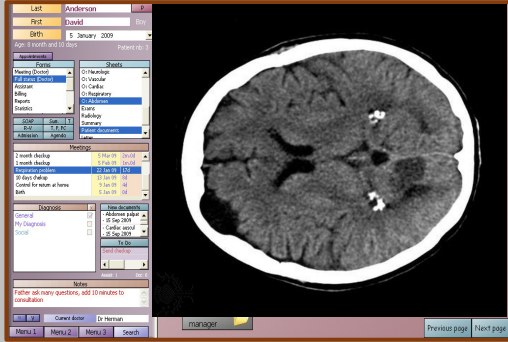
Capability $\neq$ Understanding

Why do we need AI?

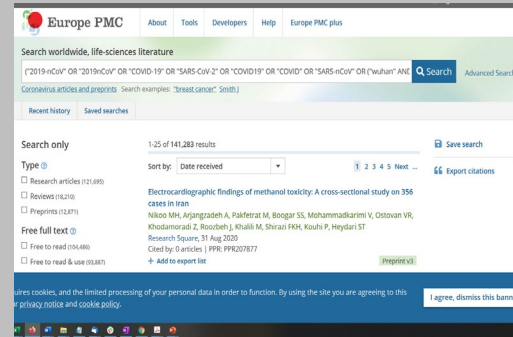
- Climate change
- Biodiversity loss
- Global health crises
- ...

Medical/Life science data is abundant...

Patient journals



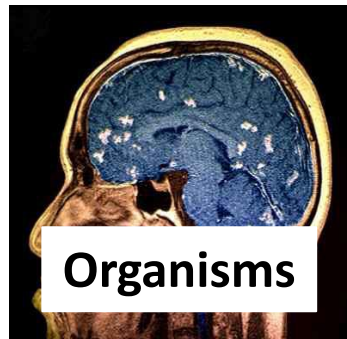
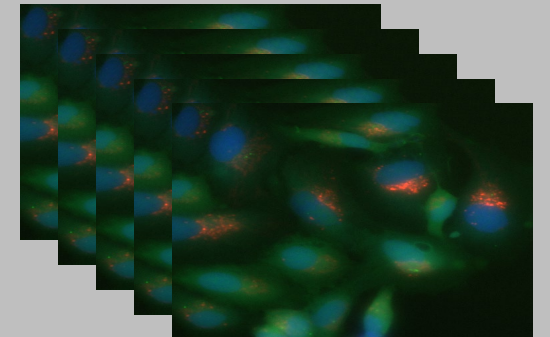
Scientific literature and other texts



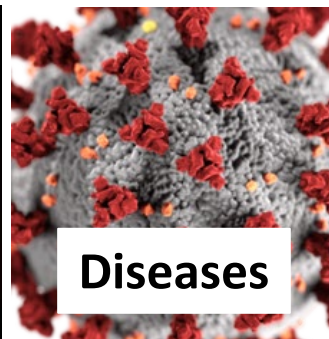
Databases



Large unstructured research datasets



Organisms



Diseases



Therapies



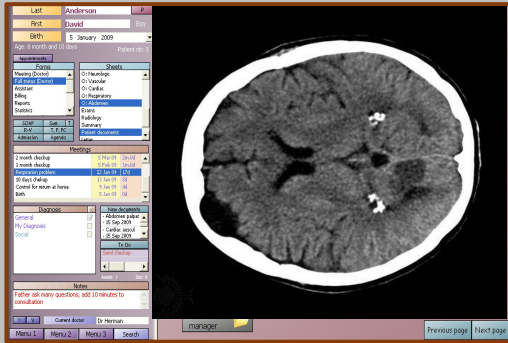
Ecosystems



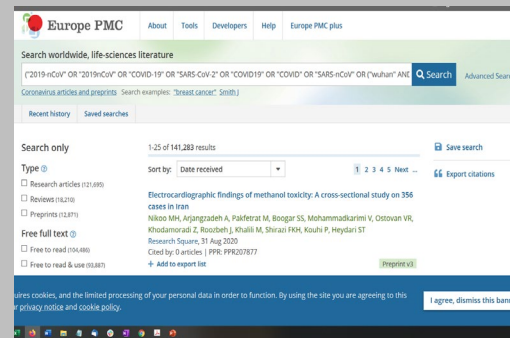
Policies

...but scattered and very complex

Patient journals



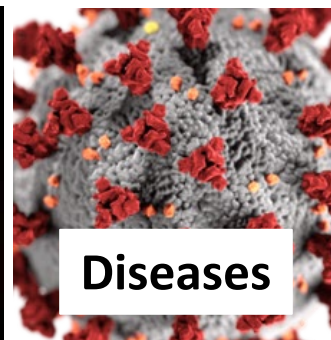
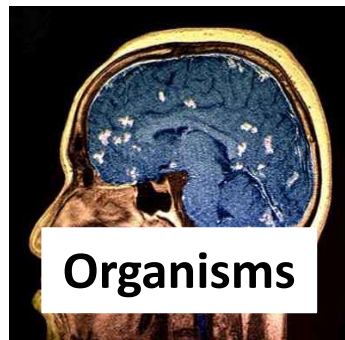
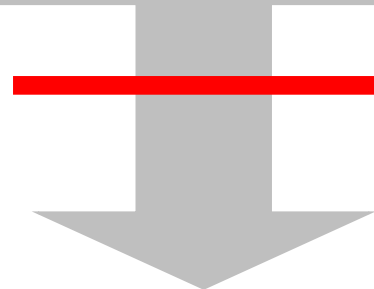
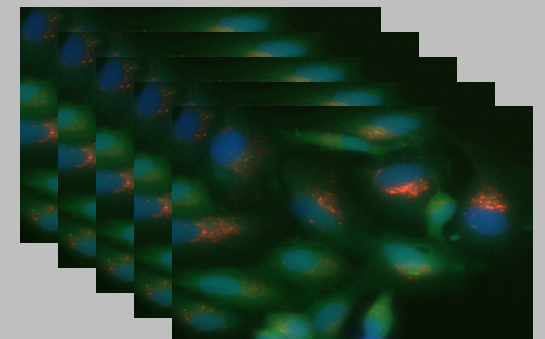
Scientific literature and other texts



Databases

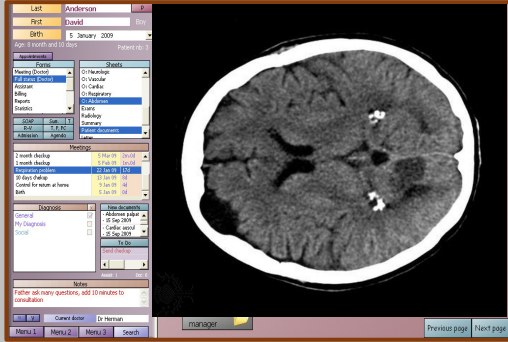


Large unstructured research datasets

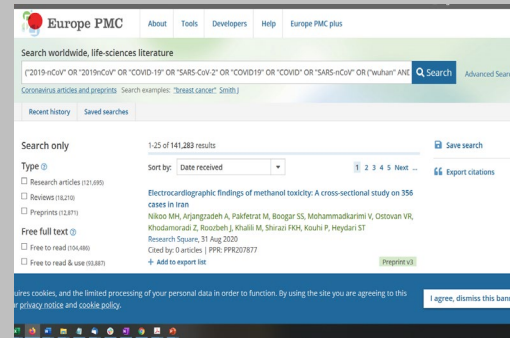


...but scattered and very complex

Patient journals



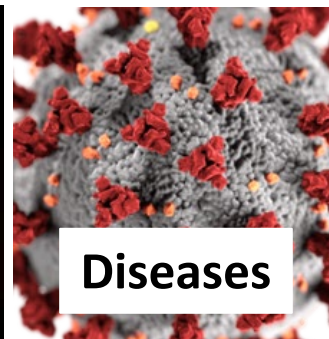
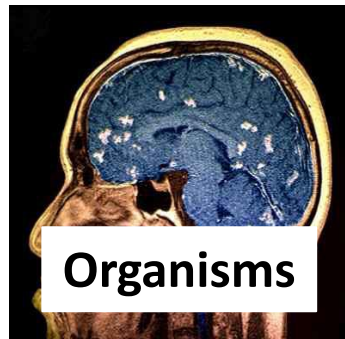
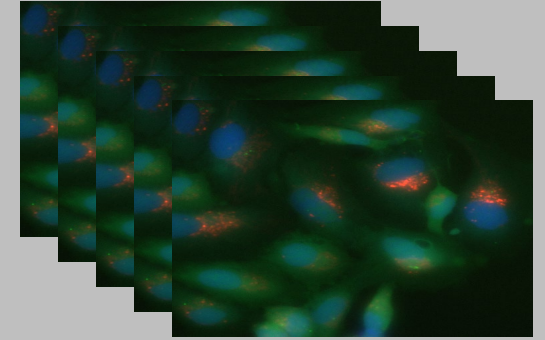
Scientific literature and other texts



Databases



Large unstructured research datasets



There are many reasons we need AI

- Information scale & scatteredness
- System complexity
- High costs
- Staff shortages
- Human variability, biases and errors
- Harm to humans/animals

AI development

Define
problem

Obtain
data

Prepare
data

Train
model

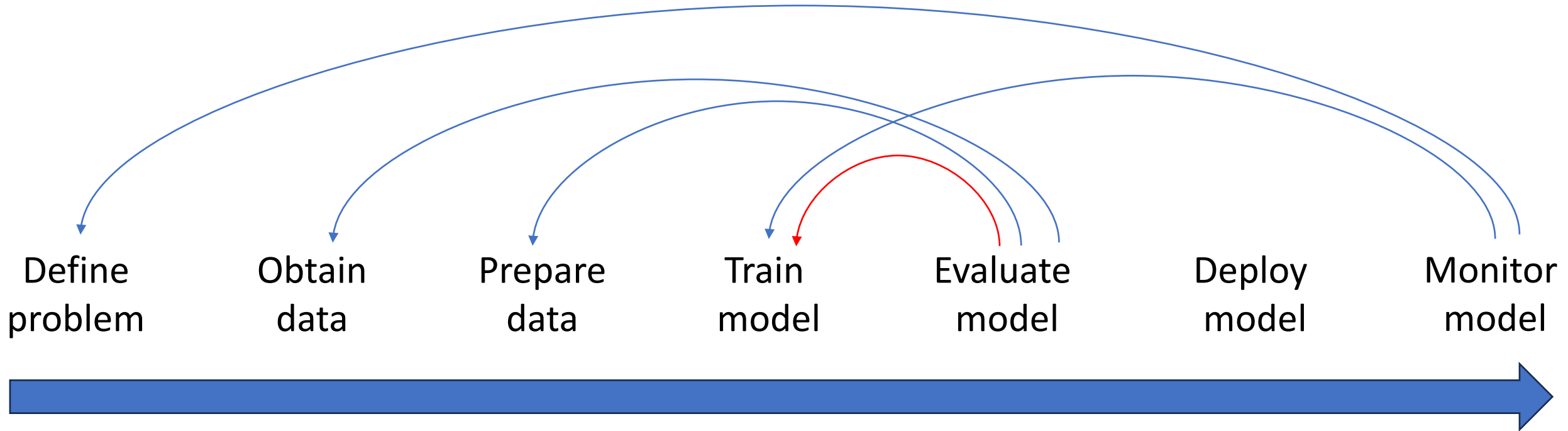
Evaluate
model

Deploy
model

Monitor
model



AI development is an iterative process



Setting clear goals

Bad

AI model for diagnosing
pancreatic cancer

Good

AI model for classifying CT images
from XYZ machine as pancreatic
cancer vs healthy with 99%
sensitivity and 95% specificity

What are deep neural networks, and how are they trained?

How we perceive the world

Sensitive detection of lysosomal membrane permeabilization by lysosomal galectin puncta assay

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Keywords: breast involution, *C. elegans*, cell death, galectin, lysosome

Abbreviations: AF, Alexa Fluor; *C. elegans*, *Caenorhabditis elegans*; BAX, BCL2-associated X protein; CD63, CD63 molecule; CTSB, cathepsin B; DPBS, Dulbecco's phosphate-buffered saline; EEA1, early endosome antigen 1; EGFP, enhanced green fluorescent protein; ER, endoplasmic reticulum; GOLGA1, golgin A1; HRP, horseradish peroxidase; HSPA9, heat shock 70kDa protein 9 (mortalin); L4, stage 4 larvae; LAMP, lysosomal-associated membrane protein; LGALS, lectin galactoside-binding soluble (galectin); LLOMe, L-leucyl-L-leucine methyl ester; LMNB1, lamin B1; LMP, lysosomal membrane permeabilization; MAP1LC3B, microtubule-associated protein 1 light chain 3 β ; MOMP, mitochondrial outer membrane permeabilization; NUC-1, abnormal NUClease (*C. elegans* homolog of DNase II); PDIA2, protein disulfide isomerase family A, member 2; RAB5, RAB5, mem-

Lysosomal membrane permeabilization (LMP) contributes to tissue involution, degenerative diseases, and cancer therapy. Its investigation has, however, been hindered by the lack of sensitive methods. Here, we characterize and validate the detection of galectin puncta at leaky lysosomes as a highly sensitive and easily manageable assay for LMP. LGALS1/galectin-1 and LGALS3/galectin-3 are best suited for this purpose due to their widespread expression, rapid translocation to leaky lysosomes and availability of high-affinity antibodies. Galectin staining marks individual leaky lysosomes early during lysosomal cell death and is useful when defining whether LMP is a primary or secondary cause of cell death. This sensitive method also reveals that cells can survive limited LMP and confirms a rapid formation of autophagic structures at the site of galectin puncta. Importantly, galectin staining detects individual leaky lysosomes also in paraffin-embedded tissues allowing us to demonstrate LMP in tumor xenografts in

Introduction

Lysosomes serve as cellular recycling centers for cargo received mainly through autophagy and endocytosis. For this purpose, they are filled with hydrolases capable of degrading most cellular macromolecules. Consequently, lysosomal membrane permeabilization (LMP) and the subsequent leakage of lysosomal hydrolases into the cytosol can lead to so-called "lysosomal cell death," which can present with necrotic, apoptotic or apoptosis-like features depending on the extent of the leakage and the cellular context, and is often misnamed as autophagic cell death due to the accumulation of autophagosomes.¹⁻⁶ Lysosomal cell death is highly conserved in evolution from yeast, roundworm, and fruit fly to mammals and has important physiological functions, e.g. in mammary gland involution and immune tolerance.^{7,9} It also contributes to pathologies of various degenerative and bacterial diseases as well as efficacies of old and new cancer therapies.

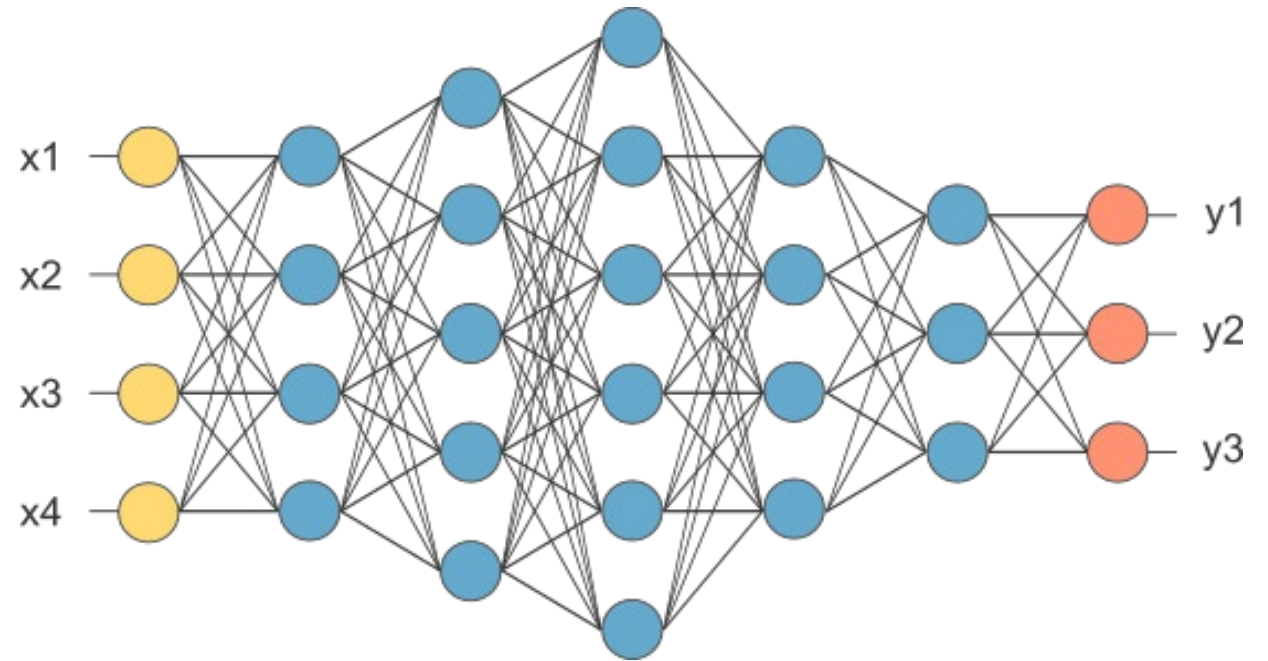
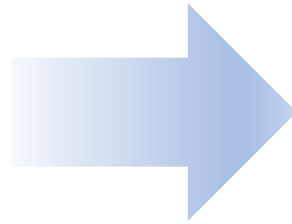
How computers perceive the world

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3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

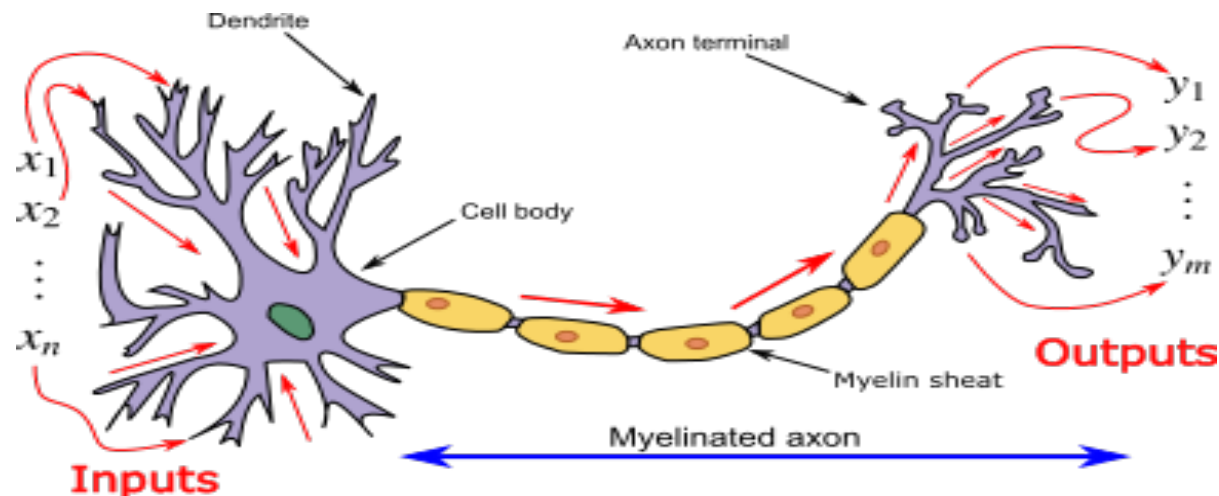
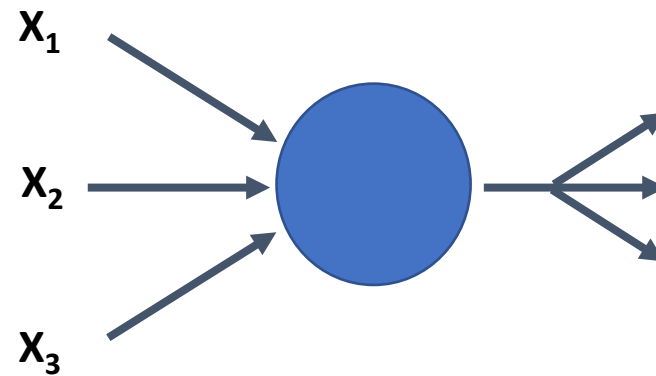
→ Data needs to be converted to a numerical representation

Deep neural networks are chains of interconnected mathematical operations

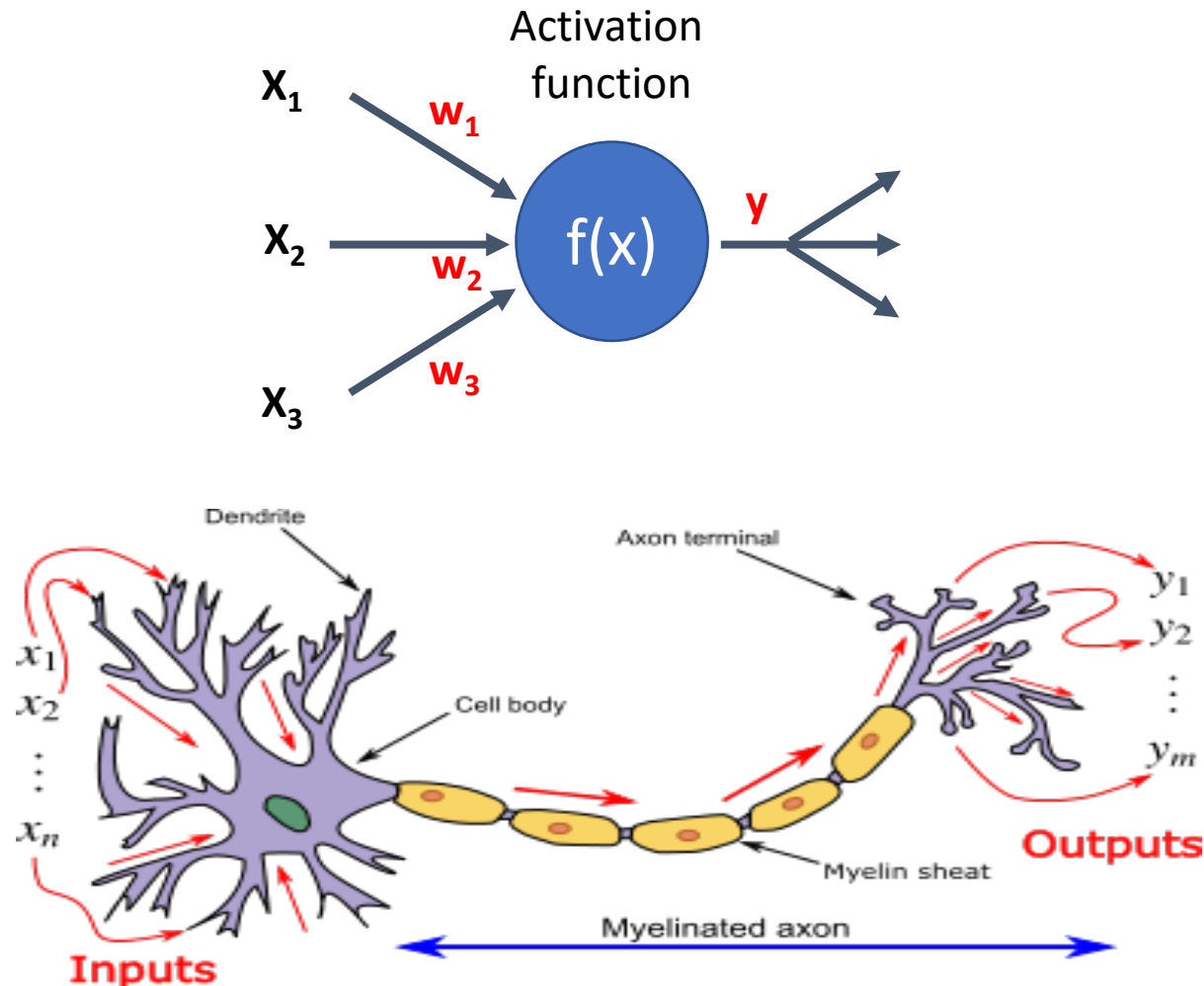
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4	3	2	1	0	3	2	5	4
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8	7	6	5	4	3	2	1	0



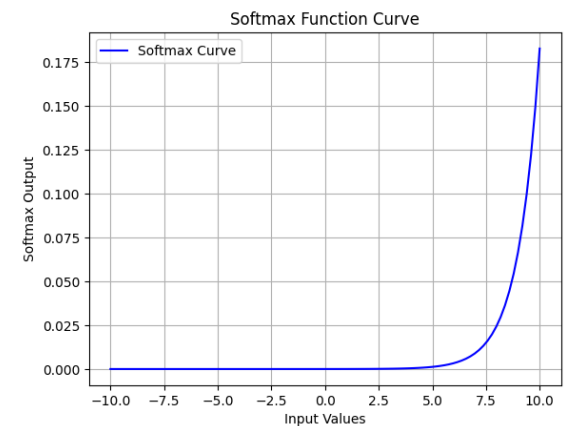
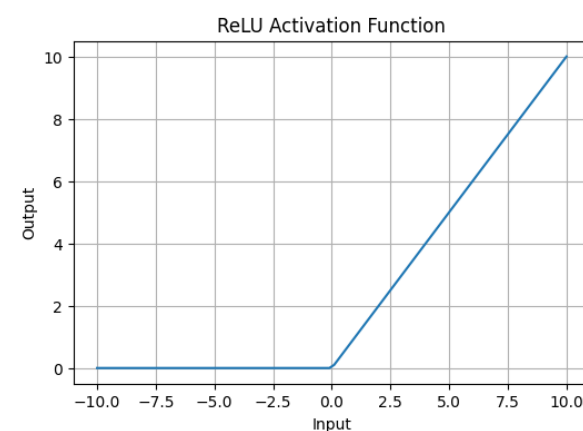
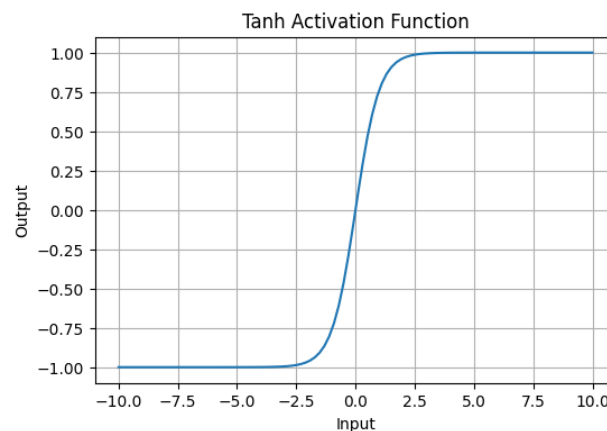
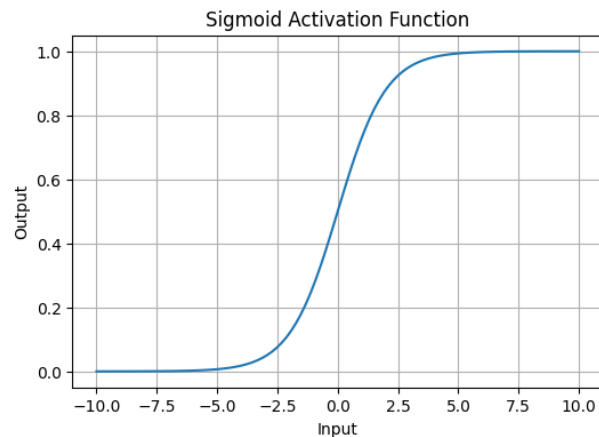
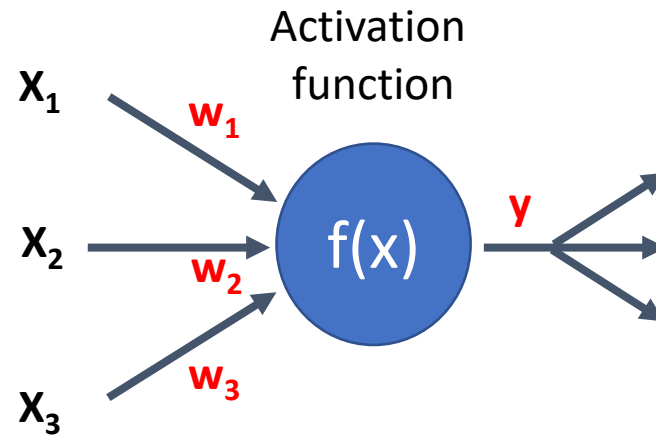
Each network node can have several input and output connections



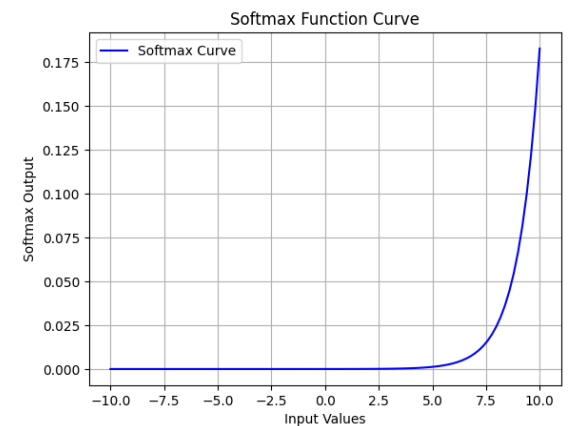
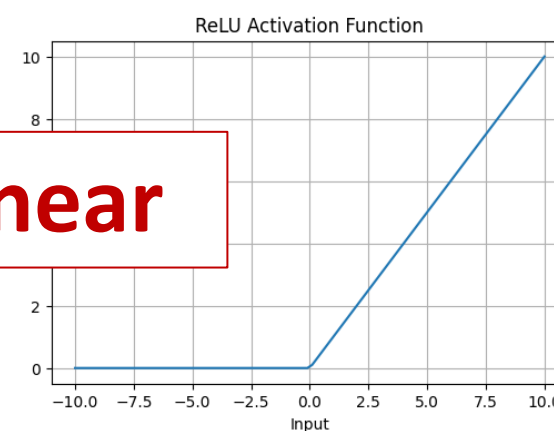
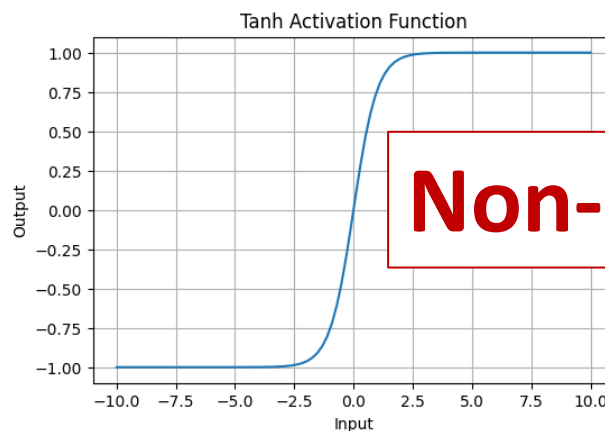
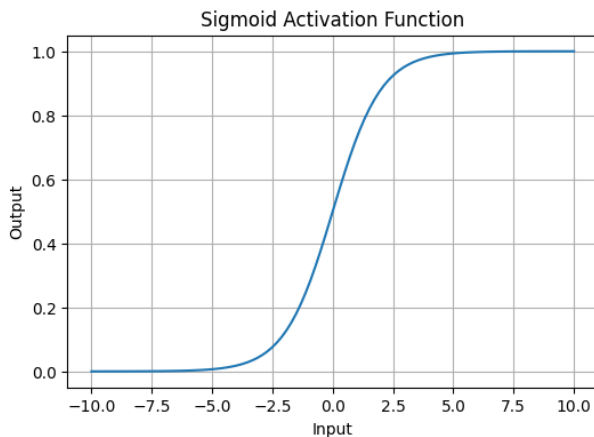
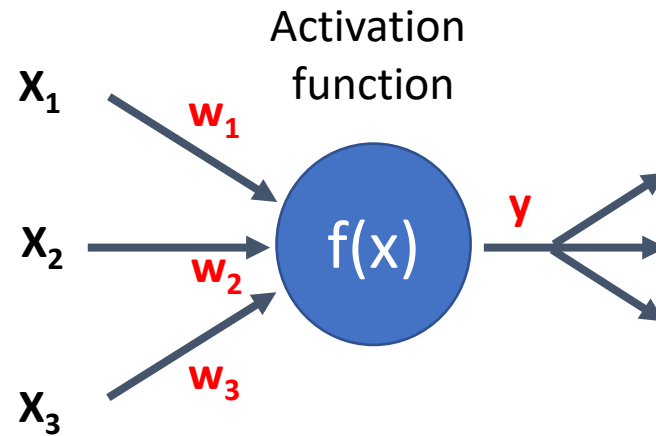
In each node inputs are weighted, aggregated and transformed to an output



Activation functions are a crucial part of the network architecture



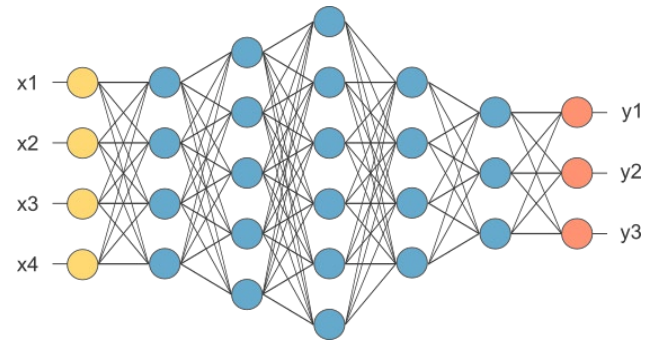
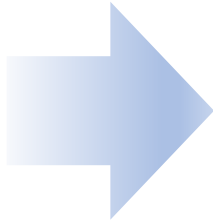
Activation functions are a crucial part of the network architecture



Non-linear

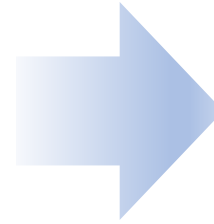
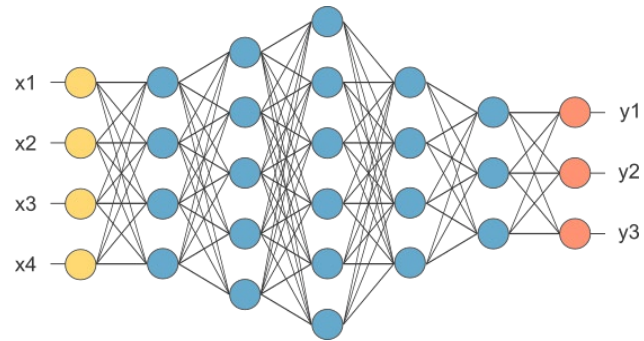
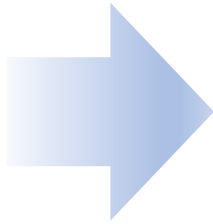
How to train a neural network...

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5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0



How to train a neural network...

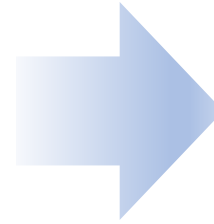
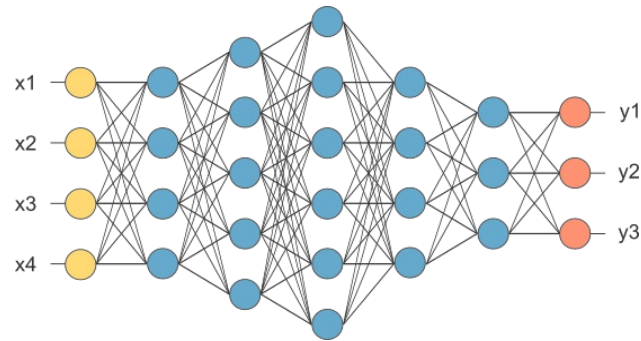
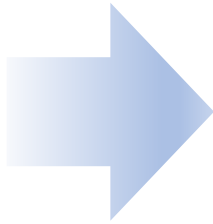
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4	3	2	1	0	3	2	5	4
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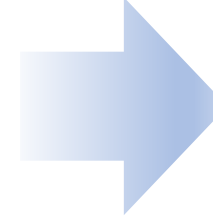
Output

How to train a neural network...

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3	0	1	2	3	4	5	6	7
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Output



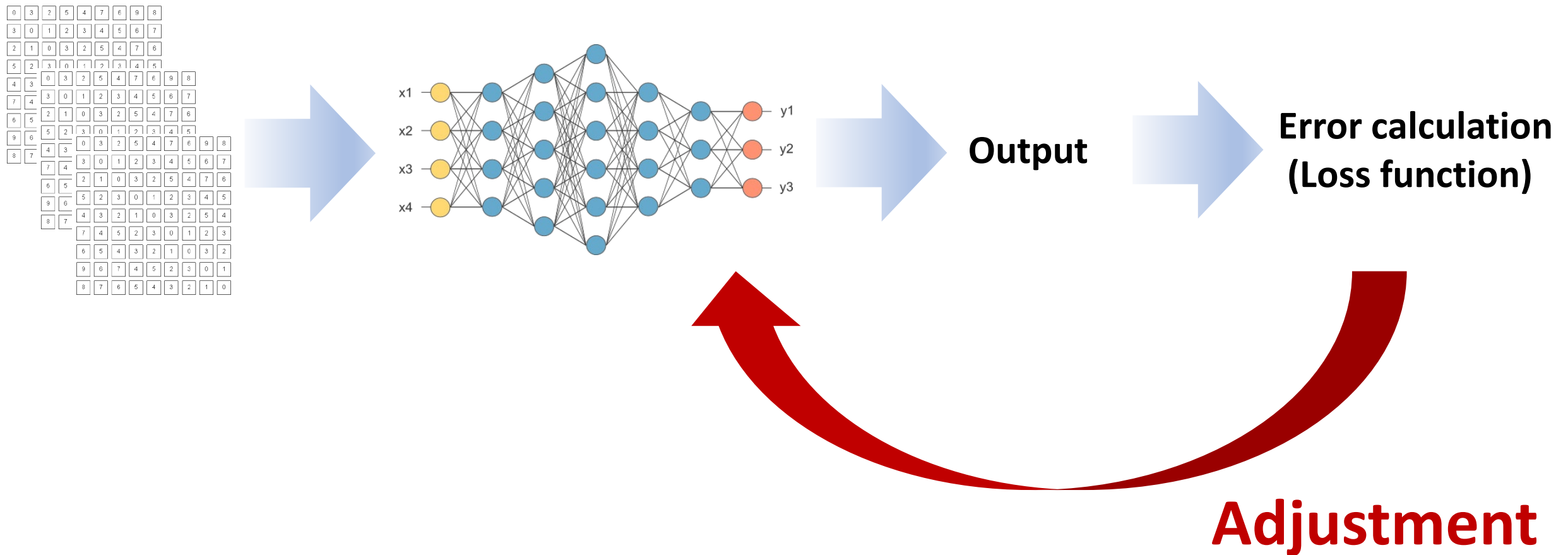
**Error calculation
(Loss function)**

Loss function is chosen depending on the task

- Classification: Binary Cross-Entropy, Categorical Cross Entropy
- Regression: Mean Squared Error (MSE), Mean Absolute Error (MAE)
- ...

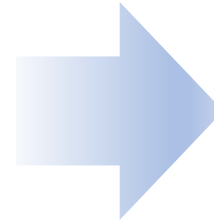
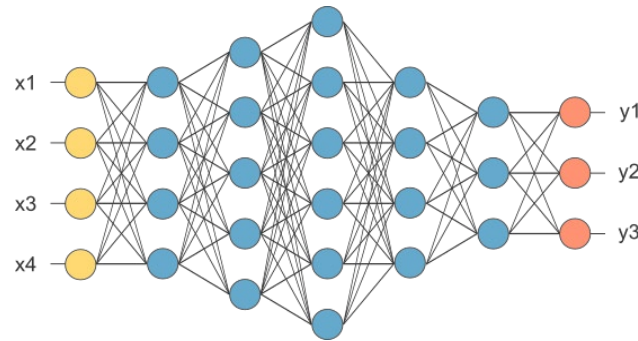
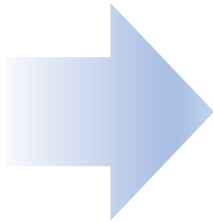
Loss function = measure of success

Weights and other parameters are adjusted to reduce errors

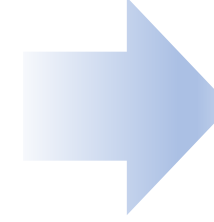


Weights and other parameters are repeatedly adjusted to reduce errors

0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
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8	7	6	5	4	3	2	1	0



Output



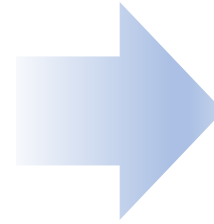
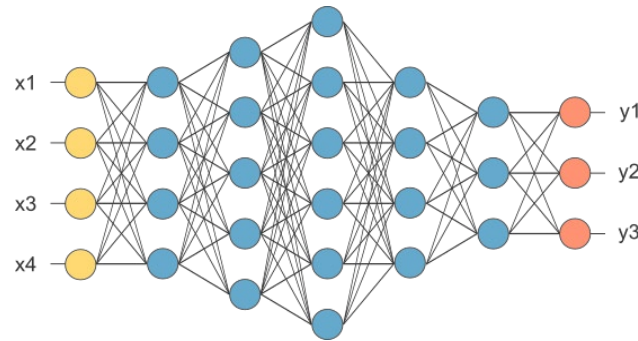
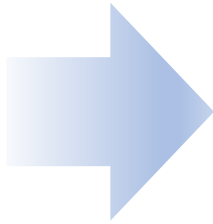
**Error calculation
(Loss function)**



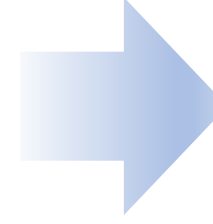
Adjustment

Weights and other parameters are repeatedly adjusted to reduce errors

0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
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6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0



Output

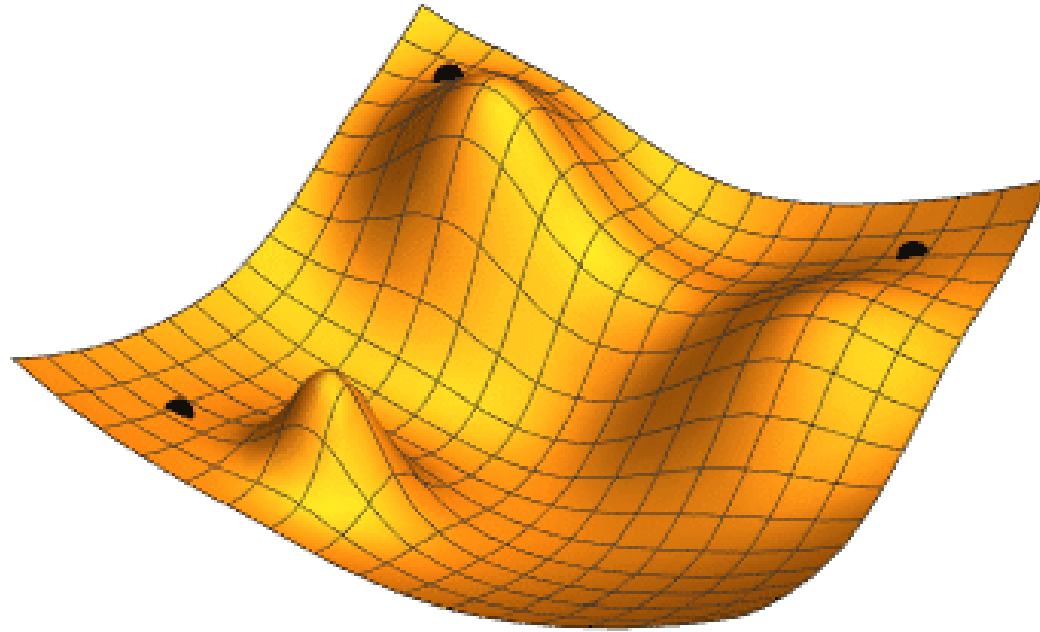


**Error calculation
(Loss function)**



Adjustment

Parameters are adjusted by gradient descent



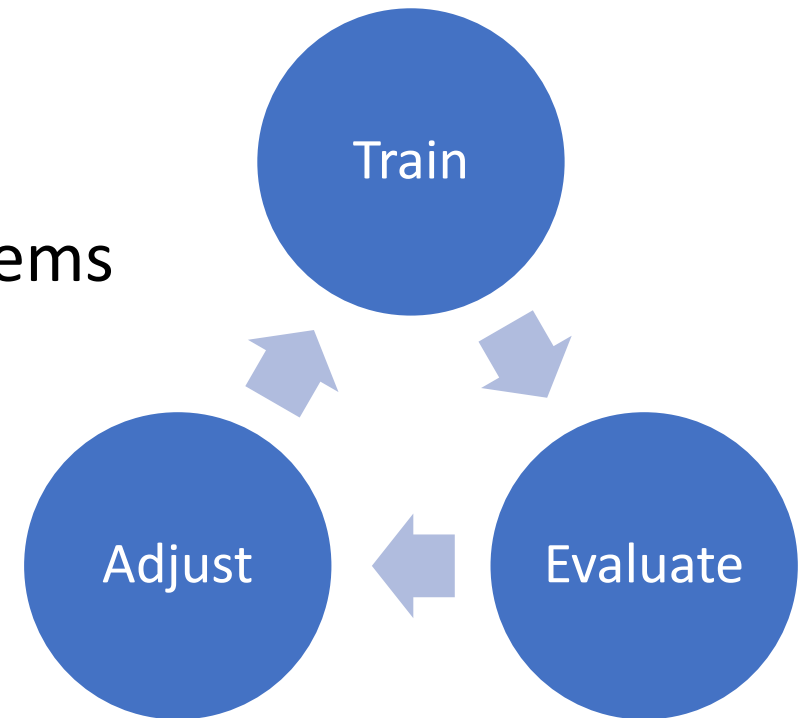
How to find the best AI model/system

AI model = the learned computational structure

AI system = integration of model(s) into a larger structure (data, processing pipelines, data, etc.)

Simple first: baseline

- Test multiple architectures for models and systems
- Vary hyperparameters
- Gradual optimization



AI can be trained in multiple ways

Supervised learning

Labeled data (x,y)

Defined target

Aim: mapping function $x \rightarrow y$

Common uses: regression,
classification

Unsupervised learning

Unlabeled data

No defined target

Aim: Learn structure of the data

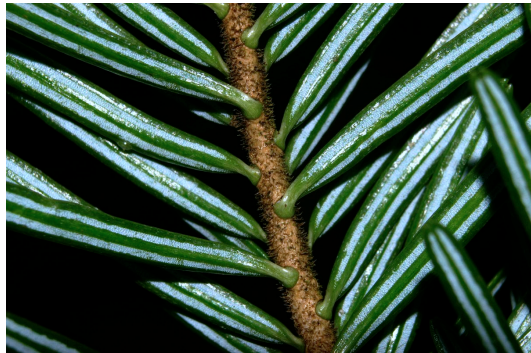
Common uses: clustering,
dimensionality reduction

Supervised learning



Oak

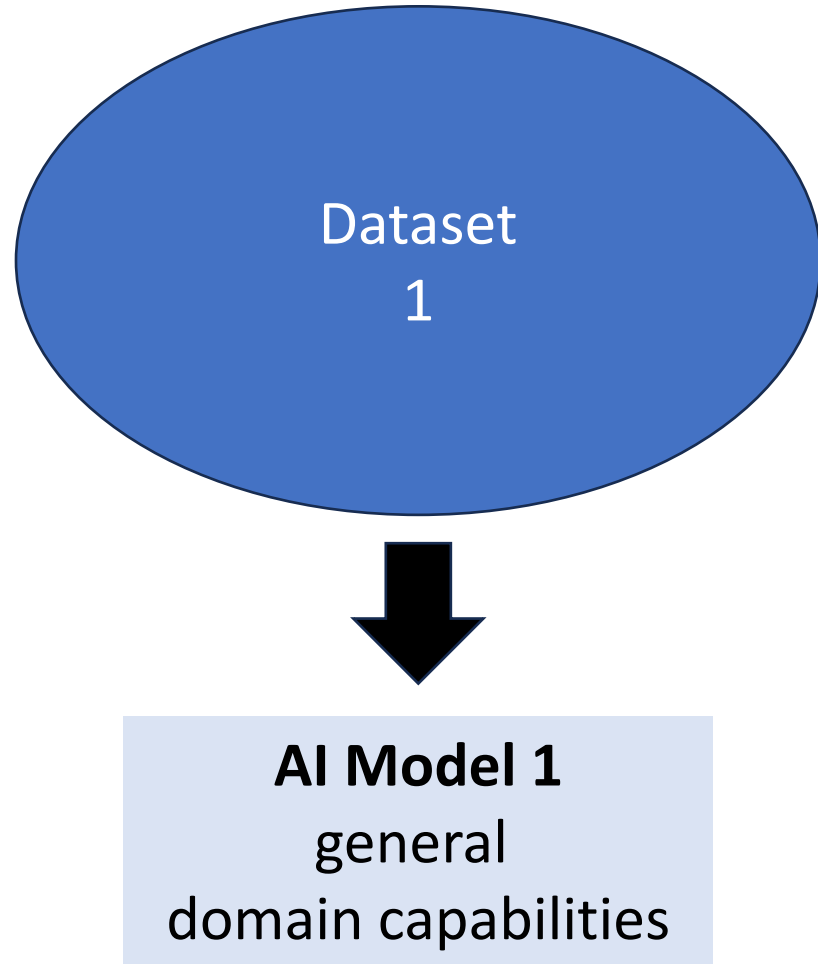
Unsupervised learning



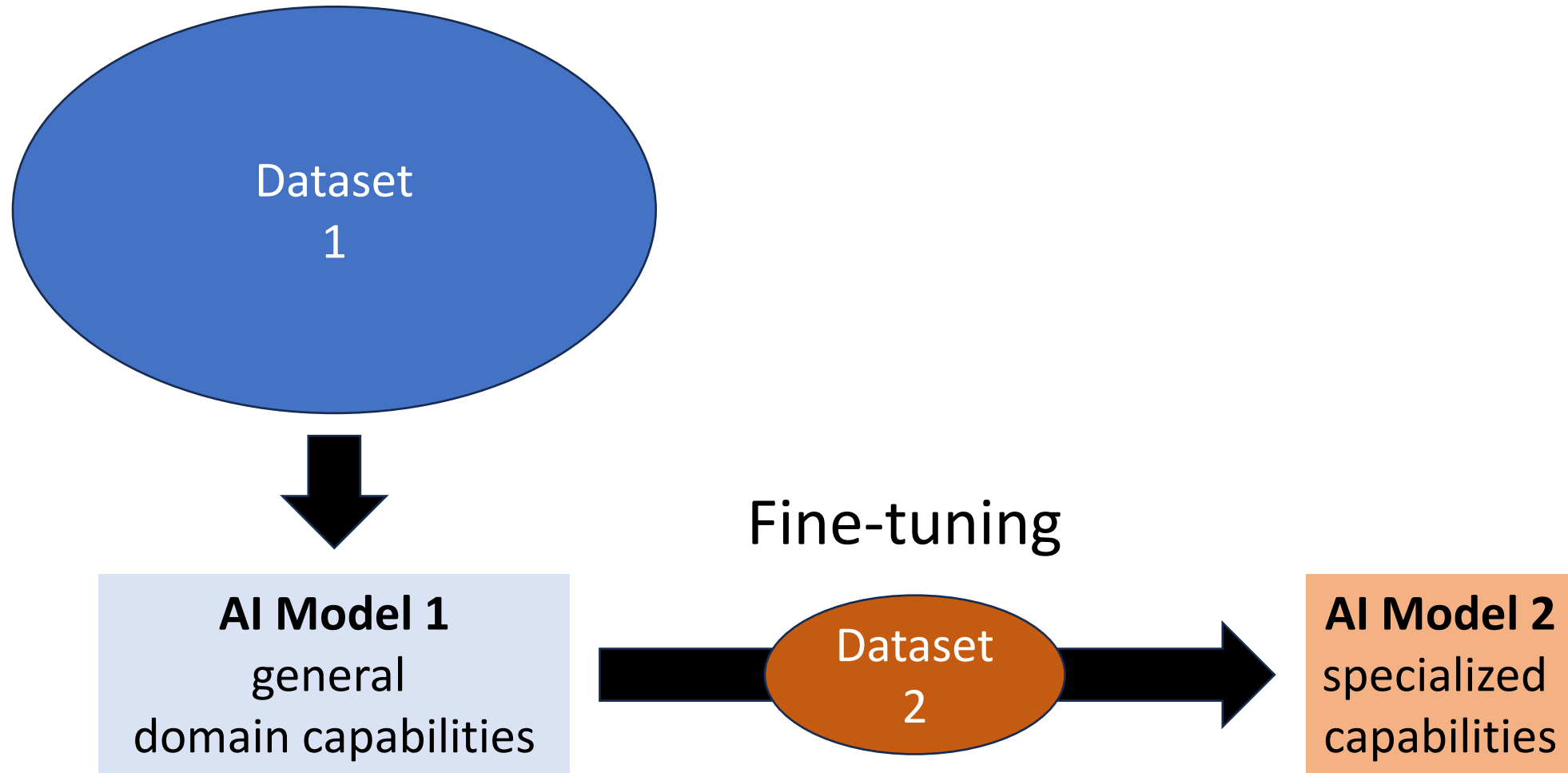
Other learning paradigms

- **Semi-supervised learning:** learning from a mix of labeled and unlabeled data
- **Self-supervised learning:** learning from unlabeled data using pretext tasks for supervision
- **Reinforcement learning:** learning by rewarding/penalizing actions

Transfer learning enables more efficient AI training



Transfer learning enables more efficient AI training



Ultimate training goal

Train model that performs well on previously **unseen data**

→ Model can **generalize**

To evaluate performance datasets are split

- **Training set:** to train models
- **Validation set (Development set):** to evaluate trained models during development
- **Test set:** to assess generalization after development

Train	Validate	Test
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- Train, validation and test set must be independent (no related examples)
- Similar data distribution in train, validation and test set
- Balance sample groups

To evaluate performance datasets are split

- **Training set:** to train models
- **Validation set (Development set):** to evaluate trained models during development
- **Test set:** to assess generalization after development
- Additional independent test data is desirable

Train	Validate	Test
		Independent test data

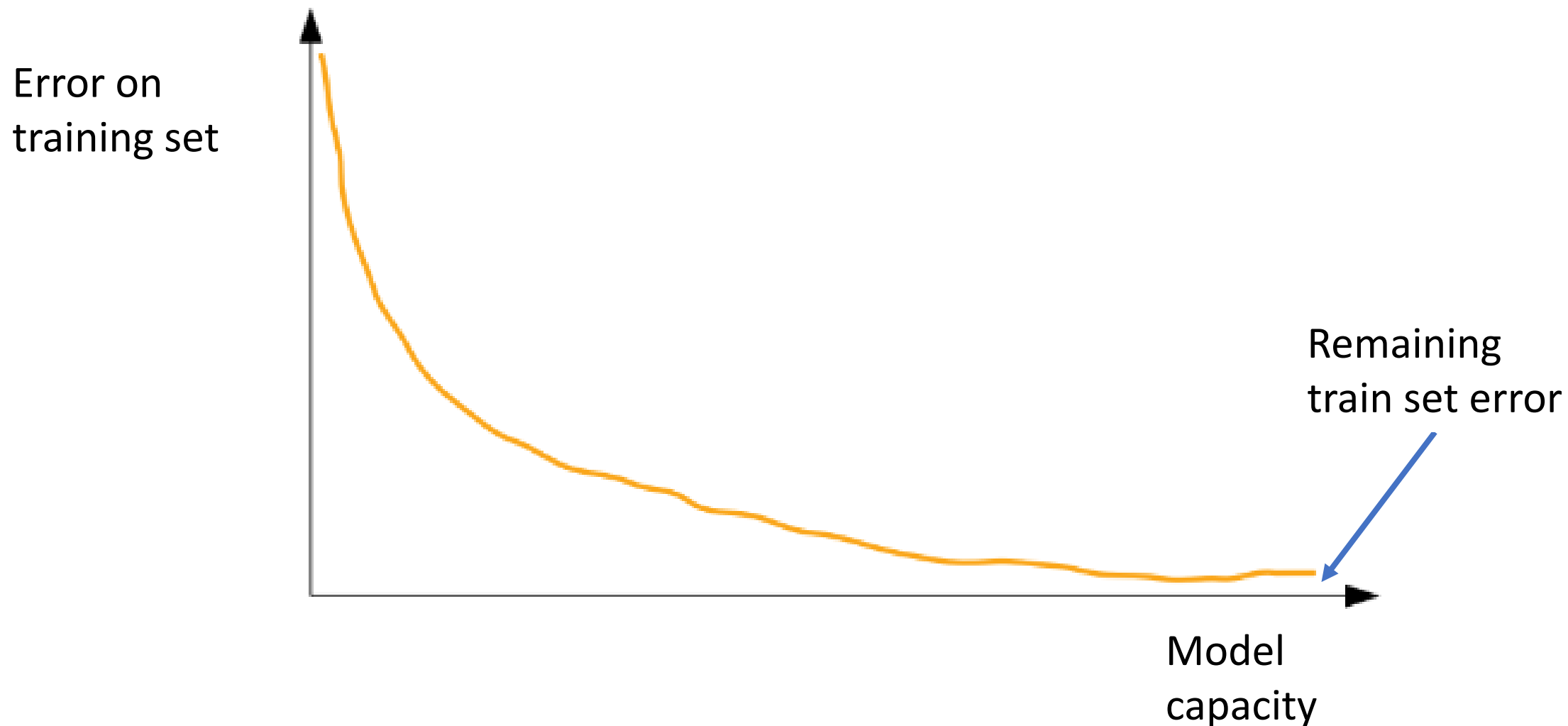
K-fold cross validation can be used when datasets are small

Train	Train	Train	Validate	Test
Validate	Train	Train	Train	Test
Train	Validate	Train	Train	Test
Train	Train	Validate	Train	Test
				Independent test data

Splitting is not always easy - avoid information leakage!!!

- 4 patient groups, 5 patients each
- 3 time points per patient
- 2 recordings taken each time point from same patient

Training goal: Minimizing the training set error



Model capacity is dependent on network architecture

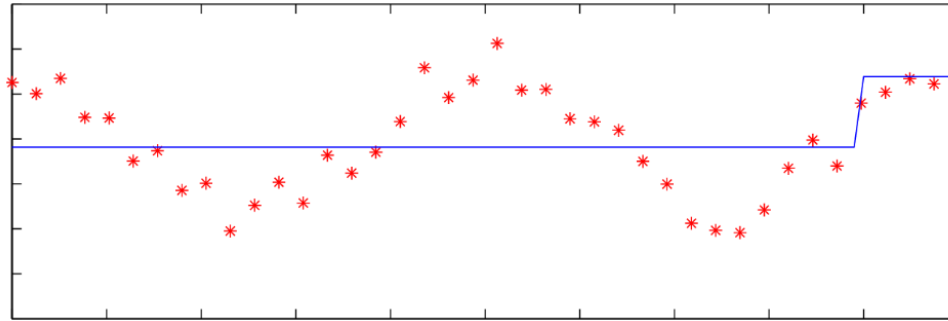
- Number of hidden layers
- Number of nodes/layer
- Network components

→ “**Hyperparameters**”: optimized by experimentation

Increasing model capacity improves data fitting but can eventually lead to overfitting

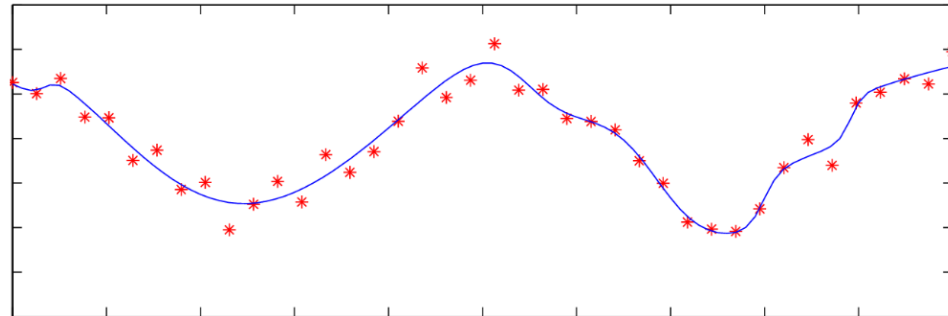
Model capacity

Low



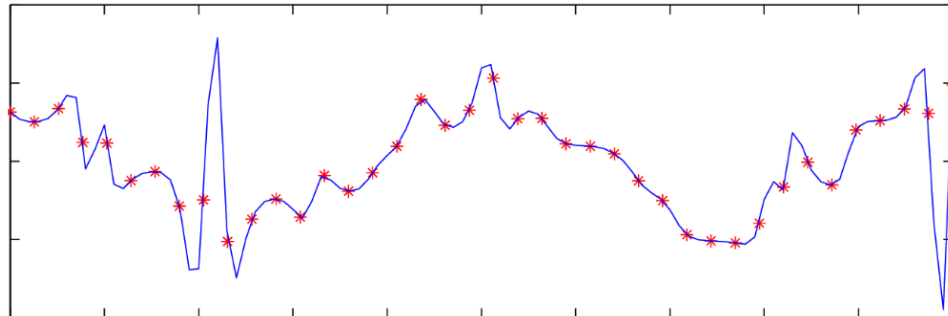
Underfitting

Medium



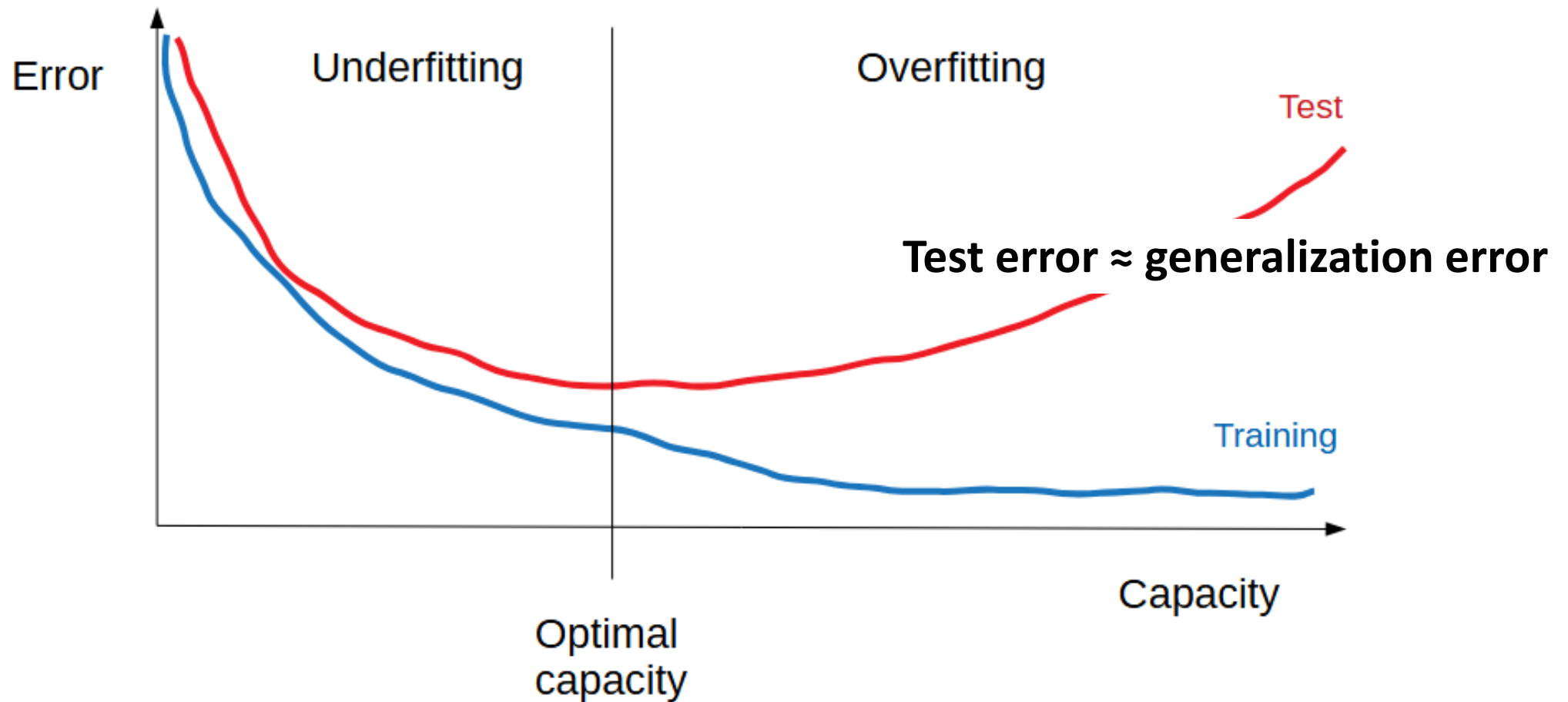
Good fit

High

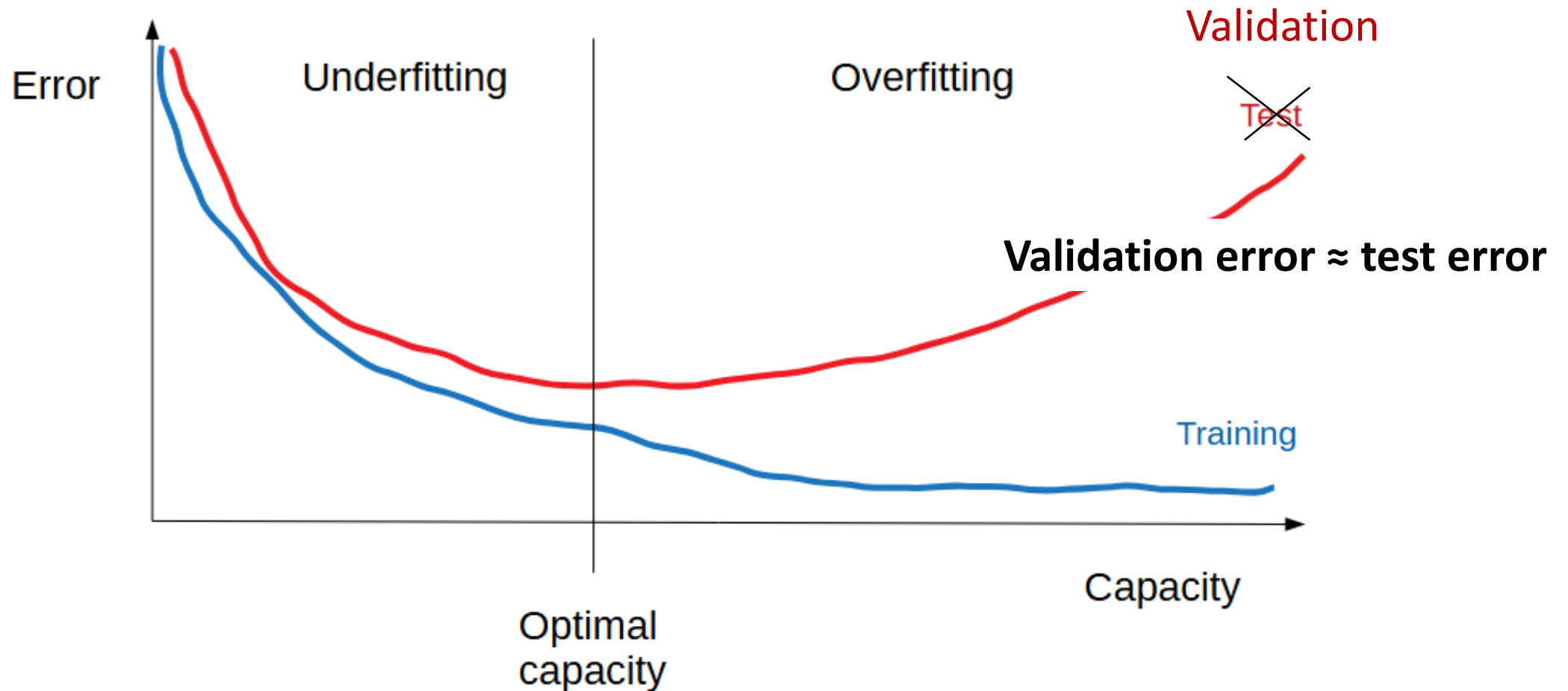


Overfitting

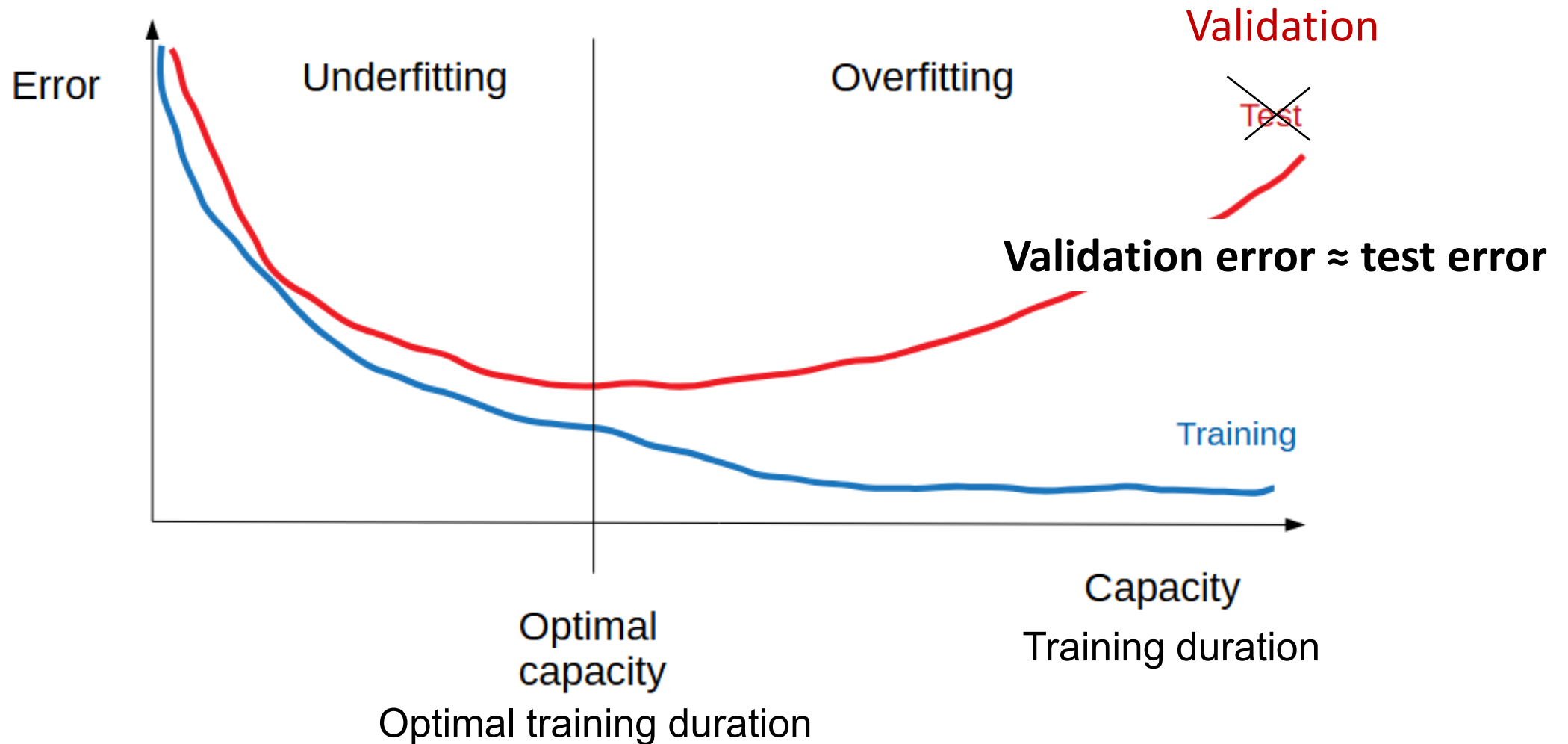
Training goal: Minimizing test error



Training goal: Minimizing validation error



Training goal: Minimizing validation error



Summary

- Deep learning relies on deep neural networks that learn from examples rather than explicit instruction
- NLP is the subdomain of AI that concerns human language
- Models can be specialized or more general-purpose (but not yet AGI)
- AI development is an iterative process
- Training relies on the repeated adjustment of network parameters to minimize the loss function; there are several learning paradigms which can also be combined
- Generalization must be evaluated on unseen data and can be compromised by overfitting